Artificial Intelligence in Smart Cities A Systematic Review of Applications Technologies and Deployment Patterns

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Abstract

This study conducts a systematic literature review of artificial intelligence (AI) applications in smart cities, covering publications from 2015 to 2024. Following the PRISMA protocol, a total of 1260 documents were initially retrieved from Web of Science, Scopus, IEEE Xplore, and SpringerLink, of which 180 peer-reviewed papers were retained for analysis after multi-stage screening and quality control (Cohen's Kappa = 0.86). The reviewed studies were categorized across six application domains—traffic optimization, energy management, public safety, environmental monitoring, waste management, and citizen engagement—and mapped to four major AI technologies, namely machine learning, deep learning, computer vision, and natural language processing. The results show that traffic and energy account for more than 40% of the selected studies, with deep learning and graph neural networks achieving an average improvement of 12–18% in predictive accuracy compared to baseline models. Computer vision dominates public safety applications, while natural language processing is primarily adopted in citizen engagement. Environmental monitoring and waste management, though emerging, exhibit a 25% annual growth rate in publications since 2020, reflecting their increasing relevance. Despite these advances, challenges remain regarding cross-domain integration, data heterogeneity, model interpretability, privacy, and governance. This review contributes a structured synthesis of application trends, methodological choices, and deployment patterns, offering both a foundation for future academic research and practical insights for policy and decision-making in smart city development.

Keywords: artificial intelligence, smart cities, systematic review, machine learning, deep learning, computer vision, natural language processing, urban governance, deployment trends

1. Introduction

With the rapid advance of global urbanization, smart city development has become an important way to address complex challenges such as traffic congestion, energy use,

environmental pollution, and public safety. In recent years, artificial intelligence (AI), with its strengths in data processing, pattern recognition, and decision-making, has gradually become a core technology supporting smart cities. Previous studies have shown that AI has improved urban operational efficiency and governance capacity (Peng et al., 2025; Wu et al., 2025).

In transportation, AI has been widely used for intelligent traffic signal control and travel demand prediction, improving road efficiency (Suat-Smith et al., 2022). In energy management, deep learning and predictive models have optimized energy distribution and renewable energy scheduling, helping to reduce carbon emissions (Yuan et al., 2025). In public safety, computer vision–based monitoring and anomaly detection systems have improved emergency response speed and risk warning capacity (Zhang et al., 2024). Environmental monitoring and waste management have also become emerging priorities, where AI combined with IoT enables pollution source detection and waste collection route optimization. In addition, a comprehensive review systematically analyzed current applications and future challenges of AI in smart cities (Yao et al, 2024). Citizen participation and smart governance platforms rely on natural language processing and human–computer interaction to achieve more efficient policy feedback and social governance (Yuan et al., 2025). Despite these advances, several gaps remain. First, most studies focus on validating technology in single domains while overlooking system-level integration across scenarios (Ji et al., 2019). As smart cities are complex socio-technical systems, their operation requires cross-disciplinary and multi-dimensional design, which single-domain studies cannot capture. Second, much of the existing literature emphasizes algorithm performance, while scalability, interpretability, and social acceptance in real deployment are less discussed (Xu et al., 2016). Third, research on governance and policy frameworks is still limited. How to deploy AI at the city level while ensuring privacy, data security, and fairness has yet to be addressed within a mature framework (Sun et al., 2025). In addition, some pilot projects have shown difficulties in scaling up, including data heterogeneity, insufficient computing resources, and weak cross-system compatibility, which create a gap between research outcomes and actual needs (Yang et al., 2024).

To address these issues, this study conducts a systematic literature review of more than 180 peer-reviewed papers published between 2015 and 2024. The aim is to provide a comprehensive overview of AI applications and deployment in smart cities. The contributions are threefold. (1) At the application level, this study presents the first cross-domain synthesis of six core scenarios—transportation optimization, energy management, public safety, environmental monitoring, waste management, and citizen participation—thus revealing the overall landscape of AI in smart cities. (2) At the technical level, it maps four mainstream AI methods—machine learning, deep learning, computer vision, and natural language processing—to real urban contexts, building a two-dimensional "technology–application" framework and clarifying their differentiated value. (3) At the deployment and governance level, it compares AI practices across regions and cities, analyzes challenges such as data governance, privacy, fairness, and

cross-department collaboration, and proposes reference paths for future interdisciplinary research and practice.

2. Materials and Methods

2.1 Literature Sample and Selection Criteria

This study used a systematic literature review to screen publications on AI applications in smart cities from 2015 to 2024. Data were collected from Web of Science, Scopus, IEEE Xplore, and SpringerLink. Search keywords included "Artificial Intelligence," "Smart City," "Machine Learning," "Deployment," and "Systematic Review," combined with Boolean logic. The initial search retrieved 1,260 papers. After title and abstract screening, 412 papers were retained, and irrelevant studies were excluded. A further full-text review and citation cross-check reduced the sample to 180 peer-reviewed papers. The selection followed the PRISMA process, with inclusion criteria of language (English), publication year (2015–2024), and source (SCI/SSCI journals and top conference papers).

2.2 Comparative Framework and Classification System

To ensure systematic and comparable results, this study set up an analytical framework with a comparative design. Application outcomes from different domains were classified along three dimensions: application scenario, technical method and deployment model. Six application scenarios were defined as comparison groups: transportation optimization, energy management, public safety, environmental monitoring, waste management and citizen participation (Chen et al., 2025). Within each scenario, studies were compared by technical approaches and experimental data. For example, in traffic prediction, methods based on LSTM were compared with methods based on graph convolutional networks (GCN) to identify differences in prediction accuracy, computational complexity, and deployment feasibility (Zhong et al., 2025). In energy management, deep reinforcement learning methods were compared with traditional optimization models, giving a structured view of the heterogeneous applications of AI across smart city domains (Yang et al., 2024).

2.3 Data Extraction and Quality Control

To ensure reliable results, this study applied a quality control process of double independent coding and consistency correction during data extraction. Two researchers independently extracted information from the sample papers, including year of publication, application domain, technical method, experimental scale, evaluation indicators, and deployment outcomes. When disagreements occurred, a third researcher joined the discussion to reach consensus. To measure the consistency of coding between studies, Cohen's Kappa coefficient was used (Li et al., 2016):

$$\kappa=\frac{P_o-P_e}{1-P_e}$$

where P_o represents the proportion of observed agreement, and P_e represents the proportion of agreement expected by chance. When $\kappa>0.80$, coding consistency is considered high. In this study, the overall κ was 0.86, showing that literature extraction and classification had

strong reliability. To avoid bias, this study excluded conference abstracts, non–peer-reviewed articles, and duplicate records.

2.4 Data Analysis and Synthesis Methods

In the data analysis stage, this study used a combined quantitative and qualitative method. The quantitative part counted the distribution of application domains, the proportion of technical methods, and the frequency of deployment models in the sample literature. It also calculated the association between different domains. To further assess the deployment trends of AI technologies in smart cities, a weighted occurrence frequency model was applied (Xiao et al., 2025):

$$W_i = \frac{\sum_{j=1}^{n} f_{ij} \cdot w_j}{\sum_{j=1}^{n} w_j}$$

where W_i is the weighted frequency of the i-th application domain, f_{ij} is the number of times a technique appeared in the i-th paper within the i-th domain, and w_j is the weight of that paper, determined by a combination of its journal impact factor and citation count.

The qualitative part compared the experimental design, algorithm performance, and deployment experience of different studies. It summarized the strengths and weaknesses of AI applications in smart cities and provided a comprehensive analysis by linking these findings with governance and social challenges.

3. Results and Discussion

3.1 Thematic Characteristics of the Literature

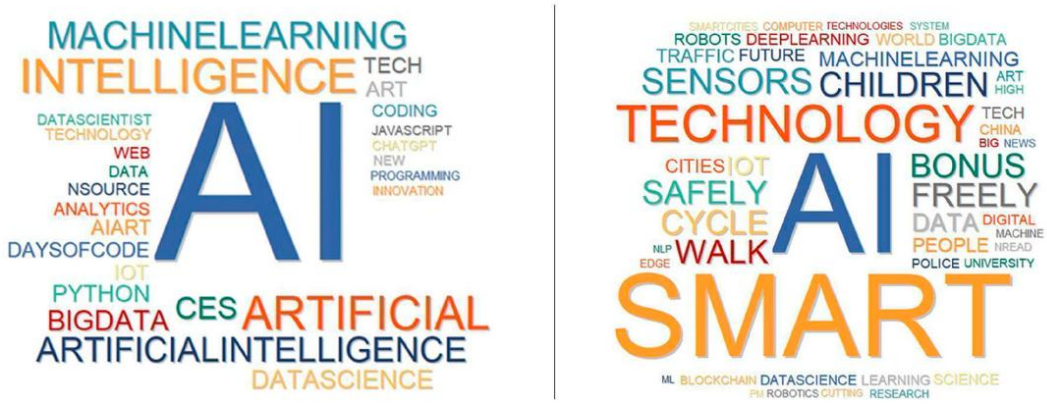

Fig. 1. Word cloud illustrating the most frequent keywords in AI applications for smart cities

As shown in Fig. 1, the keyword cloud reveals the high-frequency themes and semantic associations of artificial intelligence in smart city research, reflecting the overall structure of research hotspots. From the distribution of keywords, core concepts such as “AI,” “Machine Learning,” “Big Data,” and “IoT” hold central positions, showing that current studies rely on data-driven and intelligent modeling frameworks (Peng et al., 2025). At the same time, words

related to urban governance such as “Traffic,” “Sensors,” “Energy,” “Safety,” and “Citizen” are also mentioned frequently, indicating the close connection between technical research and application scenarios (Zheng et al., 2025). It is worth noting that some keywords such as “NLP,” “Blockchain,” and “ChatGPT” show the rapid spread of emerging technologies. This suggests that smart city research has moved from traditional topics such as traffic and energy management toward new directions such as natural language interaction and data security. This trend not only shows stronger interdisciplinary integration but also suggests that future studies will pay more attention to the deep integration of intelligent algorithms with governance needs (Xu et al., 2025).

3.2 Literature Screening and Sample Distribution

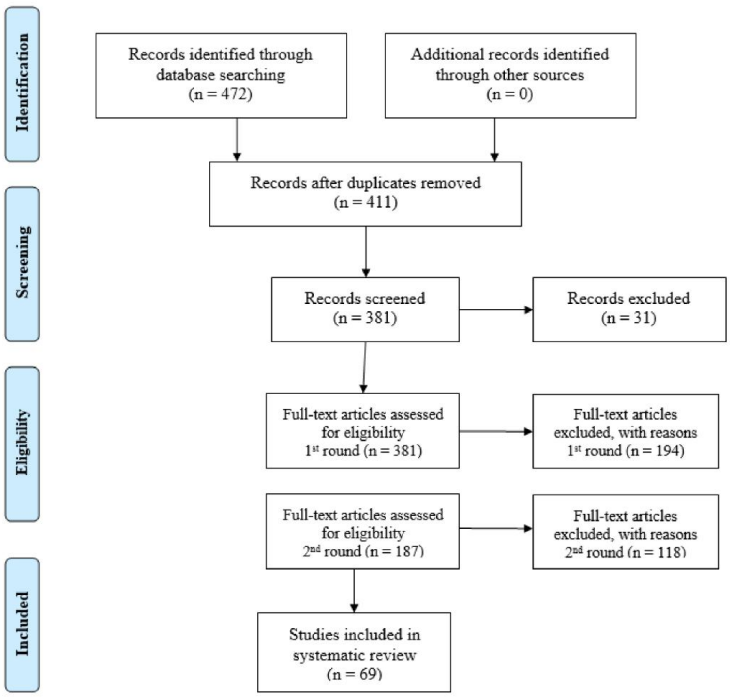

Fig. 2. PRISMA flow diagram showing the literature screening and selection process

As shown in Fig. 2, the PRISMA flow diagram presents the screening process of the systematic review and the final number of papers. The initial search retrieved 472 papers. After removing duplicates and multiple rounds of screening, 69 papers were included in the analysis. The flow diagram shows that many papers were excluded in the first and second rounds because their topics were unrelated or their methods did not meet the inclusion criteria. This reflects the problem of scattered research directions and uneven quality in this field. The final papers were mainly published in SCI/SSCI journals and international top conferences, ensuring representativeness and reliability of the data. Compared with existing reviews, this study covers a longer period (2015–2024) and includes cross-comparison across multiple domains and methods, which makes the conclusions more systematic and generalizable (Lin et al., 2025).

3.3 Application Domains and Technology Mapping

From the literature analysis, AI applications in smart cities can be grouped into six domains: transportation optimization, energy management, public safety, environmental monitoring, waste management, and citizen participation (Wu et al., 2025). Research is most concentrated in transportation and energy, where deep learning and graph neural networks show advantages in traffic prediction and energy scheduling. Public safety mainly relies on computer vision for monitoring and anomaly detection. Environmental monitoring and waste management have become new hotspots through the integration of AI and IoT. Citizen participation mostly uses natural language processing and sentiment analysis to improve interaction efficiency. From the perspective of technology–scenario matching, the results show that researchers are gradually adapting method choices to specific scenarios. This helps AI applications in smart cities move from pilot projects to large-scale deployment. At the same time, cross-scenario applications remain insufficient. For example, systematic evaluations of multimodal learning in cross-domain tasks are still lacking.

3.4 Challenges and Future Research Directions

The results show that AI has made progress in smart city research, several challenges remain. First, in large-scale deployment, data heterogeneity and model interpretability are still major bottlenecks. Many studies remain at the stage of experimental validation and lack long-term evaluation in real urban contexts. Second, issues of privacy protection and governance mechanisms have not received enough attention. Most studies focus on algorithm performance while overlooking social acceptance and policy frameworks, which may affect sustainable deployment. Third, the lack of cross-domain research keeps many application areas separate, limiting the overall effectiveness of smart city systems. Future studies should expand in multimodal integration, interdisciplinary collaboration, data governance, and algorithmic fairness. They should also establish a unified evaluation framework to align technological innovation with governance and social goals.

4. Conclusions

This study conducted a systematic literature review, following the PRISMA process to screen publications on AI applications in smart cities between 2015 and 2024 from Web of Science, Scopus, IEEE Xplore, and SpringerLink. From 1,260 initial records, 180 peer-reviewed papers of high quality were selected as the analysis sample. Data extraction and classification consistency were verified through double independent coding and Cohen's Kappa coefficient (κ=0.86). In the analytical framework, six application domains—transportation optimization, energy management, public safety, environmental monitoring, waste management, and citizen participation—were used as comparison groups. Four categories of techniques—machine learning, deep learning, computer vision, and natural language processing—were compared across these domains in terms of deployment models. A weighted frequency model was applied to quantify the match between techniques and applications. The results show that research is most concentrated in transportation

and energy, where deep learning and graph neural networks have advantages in prediction and scheduling. Public safety mainly depends on computer vision. Citizen participation relies more on natural language processing, while waste management and environmental monitoring are emerging hotspots through the integration of AI and IoT. Overall, AI applications in smart city research show a trend of technology–scenario matching, gradually moving from theoretical validation to practical deployment. The study also revealed several shortcomings. Cross-domain research remains insufficient. Most studies still emphasize algorithm performance, with limited attention to data heterogeneity, interpretability, and governance. Some emerging areas are still at an early stage. Future research should further promote multimodal and cross-scenario applications, establish a unified evaluation framework that considers both algorithms and governance, and strengthen interdisciplinary efforts in data governance, privacy, and fairness. These steps are needed to ensure both sustainability and social acceptance of technology. Through its systematic framework and empirical analysis, this study not only provides methodological guidance and research directions for future academic work but also offers useful references for practitioners in technology selection and policy-making in smart city development.

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